

24MAT110-ASSIGNMENT 2

1. Find the area of the region between the x-axis and the graph of $f(x) = x^2 - 4x + 3$ over $[0, 3]$.
2. Find the area of the region bounded by the curves $y = x$, $y = \frac{1}{\sqrt{x}}$ and $x = 2$.
3. Find the volume of the solid generated by revolving the region bounded by the curves $y = 2 \tan x$, $y = 0$ and $x = -\frac{\pi}{4}$ around the line $x = \frac{\pi}{4}$.
4. Find the arc length of $y = \left(\frac{5}{12}\right)x^{\frac{6}{5}} - \left(\frac{5}{8}\right)x^{\frac{4}{5}}$ from $x = 1$ to $x = 32$.
5. Find the area of the surface generated by revolving the curve $x = \sqrt{4y - y^2}$, $1 \leq y \leq 2$ about the y-axis.